Reproduction of Table I of arXiv:1708.02243 — update

S. Knapen, H. K. Lou, M. Papucci, J. Setford, “Tracking down quirks at the Large Hadron Collider”, Phys. Rev. D 96, 115015 (2017) · Table I, backgrounds, Figures 2–5 · 2000-event production samples and an EW LO sample · 15 October 2026 (replaces the summary of the 9 October report; that report is kept unchanged)

Executive summary

- **Trigger.** Colored: 207/873 = 0.237 ± 0.014 (paper 0.24). EW, matched + showered production sample: 0.178 ± 0.012 (paper 0.10). EW, LO cross-section ratio $\sigma(Q\bar{Q}j, p_T(j) > 200\ \text{GeV})/\sigma(Q\bar{Q}) = 0.096 \pm 0.001$ (MG5 3.7.3; 2.3.3: 0.096), equal to the paper's 0.10. The same LO ratio for colored is 0.142, not 0.24. The two benchmarks therefore behave as if they had been produced differently; this is an inference, not a demonstration.
- **Fiducial.** Colored 0.324 ± 0.057, EW production 0.333 ± 0.035; EW LO sample (first 2118 of 3000 events, run in progress) 0.300 ± 0.010; paper 0.28 for both.
- **Reconstruction.** Without SCT z smearing: colored 0.364 ± 0.103, EW 0.806 ± 0.050 (paper 0.35, 0.72). With the SCT z resolution the authors confirm (580 µm, Ref. [27]): 0.136 ± 0.073 and 0.226 ± 0.053; with 1.16 mm: 0.045 ± 0.044 and 0.161 ± 0.047. Only the setting without SCT z smearing reproduces the paper's numbers.
- **Figure 2.** With the paper's equation of motion (Eq. 3 + Lorentz force) |L| returns to zero at every second closest approach and all |L| peaks are equal. This is the analytic result of Li, Li, Pei and Zhang (arXiv:2002.07503, Appendix A) and is reproduced by three independent codes. The paper's Fig. 2 instead shows unequal half-oscillations (|L| = 0.19 and 0.29×10^{12} ħ at the 2nd and 4th approach); no tested definition, charge set, axis or equation-of-motion variant gives that.
- **Annihilation.** The paper argues that |L| = 0 only at large separation. In the classical model |L| ≈ 0 at every second closest approach, so this argument is not supported. Whether annihilation then matters needs a quantum estimate of the per-meeting probability; a worst-case estimate is small at the Table I benchmarks (≤ 3% loss for colored at 2 keV) and grows with Λ.
- **Backgrounds.** 0 of 9139 converted photons and 0 of 5000 pile-up events in the signal region (SCT z 580 µm), as in the paper; the 95% upper limit 3.3×10^{-4} per conversion depends only on the 9139 events, not on the cross section. Without SCT z smearing on the background, 146/9139 conversions pass.
- **Open:** EW ε_{reco} at Λ = 1–2 keV above the paper; signal strip width Δw 2× (EW) and 5× (colored) the paper's; the SCT z resolution; the unequal half-oscillations of the paper's Fig. 2.

Uncertainties in this report are approximate statistical uncertainties of binomial efficiencies, $\sigma = \sqrt{\varepsilon(1-\varepsilon)/n}$, with n the denominator of each efficiency (Section 1.4). No systematic uncertainty is included. Wilson intervals and p-values of the 9 October report are not repeated.

1. Samples, detector model and definitions

1.1 Signal samples

Sample	Generation	Events	Trigger definition
Production (colored, EW)	MG5 3.7.3 + Pythia 8.317, MLM (ickkw 1, xqcut 30 GeV, qCut 45 GeV, nJetMax 1, nQmatch 4, doShowerKt off), nn23lo1, default dynamical scale, 13 TeV; colored p p → t1 t1~ (+j), EW p p → qe+ qe− (+j); seeds 8001 / 8002	2000 each	$p_T(Q\bar{Q}) > 200$ GeV of the final-state pair
EW LO sample A	p p → qe+ qe−, LO, no shower; MG5 3.7.3 and 2.3.3	2000	— (σ only)
EW LO sample B (D41)	p p → qe+ qe− j, $p_T(j) > 200$ GeV, $\lvert\eta(j)\rvert < 5$, ickkw 0, xqcut 0, LO, no shower; MG5 3.7.3 (seed 41002) and 2.3.3	3000 (2118 analysed, run in progress)	all events ($p_T(Q\bar{Q}) = p_T(j) > 200$ GeV at LO)
EW sample B showered (D41 sensitivity)	sample B + Pythia 8.317, no matching	3000	$p_T(Q\bar{Q}) > 200$ GeV after the shower — not yet run
Colored LO A / B	p p → t1 t1~ (j), $p_T(j) > 200$ GeV, m = 1800 GeV	2000 each	— (σ only)

No generation cut other than those listed. The colored quirk is a Dirac fermion, colour triplet, charge 2/3 (param_card QNUMBERS: 2 spin states; FFV vertices); only its PDG number (1000006) is borrowed from the stop.

1.2 Detector model and the smearing formulas used

Element	Formula / value	Source
Layers	IBL 3.325, pixel 5.05 / 8.85 / 12.25, SCT 29.9 / 37.1 / 44.3 / 51.4 cm; |z| < 33.0 / 40.05 / 74.9 cm; B = 2 T along z	paper Sec. III; Ref. [27] lengths; IBL not in [27] (2008), Run-2 values
Pixel hit	pixel centre: $x_{meas} = p\cdot\text{round}(x/p)$, $p_{r\varphi} = 50$ µm, $p_z = 250$ µm (IBL) / 400 µm; RMS error $p/\sqrt{12}$ = 14 / 72 / 115 µm; neighbouring pixels merged	paper: “pixel size rather than the resolution”
SCT rφ	$(r\varphi)_{meas} = r\varphi + \sigma_{r\varphi}\cdot g$, g ~ N(0, 1), $\sigma_{r\varphi} = 34$ µm = 2 × 17 µm	paper: twice the intrinsic accuracy of [27]
SCT z	$z_{meas} = z + \sigma_z\cdot g$, $\sigma_z = 0$ (none), 0.58 mm (Ref. [27]; authors' reply), 1.16 mm (2 × 0.58, paper text)	all three reported

Element	Formula / value	Source
Beam spot	one $z_0 \sim N(0, 45\ \text{mm})$ per event shifts all hits; the plane is fitted about the true vertex	paper Sec. III A; per-event shift: our choice
Hits	first outward crossing of each layer by each quirk, $t < 25$ ns (every crossing as a sensitivity)	paper silent; our choice
Plane	$T_{ij} = \Sigma_a x_{a,i} x_{a,j} / (N-1)$; $\Delta s = \sqrt{\lambda_1}$, $\Delta w = \sqrt{\lambda_2}$; seeds $\Delta s < 0.05$ cm, $\Delta w < 1$ cm, factor 3	paper Sec. III C, Eq. 11
Signal region	2 hits in 7 layers and ≥ 1 in the 8th, $\Delta w < 1$ cm, $\Delta s < 0.01$ cm	paper Sec. IV
Backgrounds	[34] detector model (bremsstrahlung, ionisation), SCT z 580 μm; μ = 50 pile-up, Pythia 8 minimum bias	Ref. [34] App. A; values not stated for SCT z

1.3 Efficiencies (paper Eq. 12, $\varepsilon = \varepsilon_{trig} \times \varepsilon_{fid} \times \varepsilon_{reco}$); each is conditional on the previous stage

Efficiency	Numerator	Denominator
ε_{trig}	events with $p_T(Q\bar{Q}) > 200$ GeV	events after MLM matching (production); LO: σ(B)/σ(A), a cross-section ratio
ε_{fid}	events whose two quirks cross all 8 layers inside the lengths, first crossings at $t < 25$ ns	triggered events (colored: both quirk-hadrons charged); LO sample B: all events
ε_{reco}	events in the signal region	fiducial events
ε (Eq. 12)	product of the three	relative σ added in quadrature (three disjoint binomial stages)

1.4 Approximate uncertainty

$\hat{\varepsilon} = k / N$, $\sigma_{\hat{\varepsilon}} \approx \sqrt{(\hat{\varepsilon}(1-\hat{\varepsilon}) / N)}$, N = number of events entering that selection

For 0 passing events the formula gives 0; the 95% upper limit $1 - 0.05^{1/N} \approx 3/N$ is quoted instead. The LO trigger ratio is a ratio of integrated cross sections; its uncertainty comes from the MadGraph integration errors, $\sigma_r/r = \sqrt{((\delta\sigma_A/\sigma_A)^2 + (\delta\sigma_B/\sigma_B)^2)}$. The events are unweighted and independent; the same events enter all Λ points (correlated between Λ, not between samples).

2. Table I

2.1 Event counts of the production samples (first crossing)

Step	Colored: none	580 μm	1.16 mm	EW: none	580 μm	1.16 mm
after matching	873	873	873	1045	1045	1045
triggered	207	207	207	186	186	186
both charged (colored)	68	68	68	—	—	—
fiducial	22	22	22	62	62	62
plane found ($\Delta w < 1$ cm)	18	18	9	62	61	45
+ hit multiplicity	8	9	4	50	45	24
+ $\Delta s < 0.01$ cm = signal region	8	3	1	50	14	10

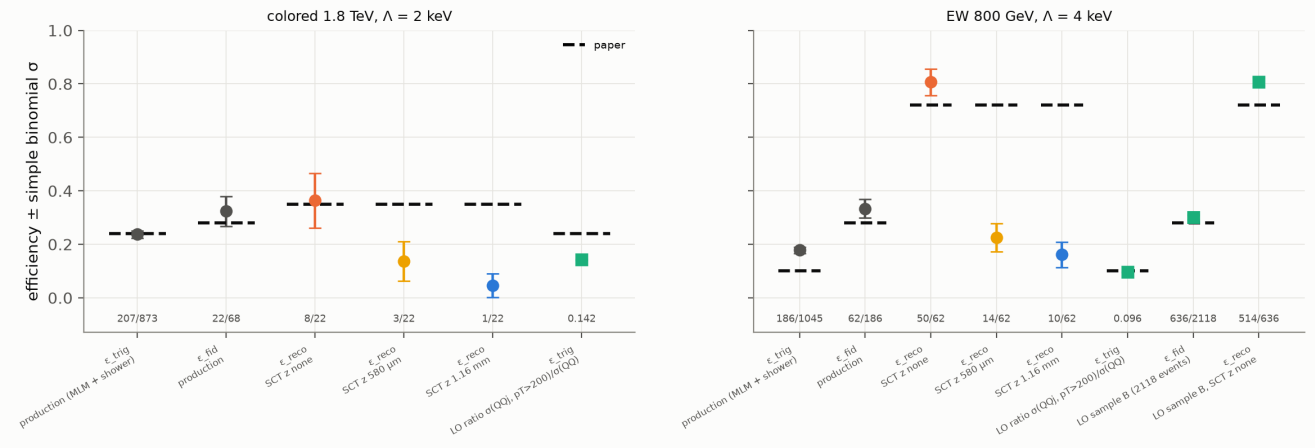
2 colored and 1 EW triggered events stopped with a numerical integration error and are counted as not fiducial.

2.2 Efficiencies against the paper

Benchmark	Efficiency	Sample / setting	k / N	ε ± σ	Paper
Colored 1800 GeV, 2 keV	ε_{trig}	production (MLM + shower)	207 / 873	0.237 ± 0.014	0.24
	ε_{trig}	LO ratio σ(QQj)/σ(QQ), MG5 3.7.3 / 2.3.3	—	0.142 ± 0.001 / 0.143 ± 0.001	0.24
	ε_{fid}	production	22 / 68	0.324 ± 0.057	0.28
	ε_{reco}	production, SCT z none	8 / 22	0.364 ± 0.103	0.35
	ε_{reco}	production, SCT z 580 μm	3 / 22	0.136 ± 0.073	0.35
	ε_{reco}	production, SCT z 1.16 mm	1 / 22	0.045 ± 0.044	0.35
	ε (Eq. 12)	production, SCT z none		0.0279 ± 0.0094	0.0235*
	ε (Eq. 12)	production, SCT z 580 μm		0.0105 ± 0.0059	0.0235*
EW 800 GeV, 4 keV	ε_{trig}	production (MLM + shower)	186 / 1045	0.178 ± 0.012	0.10
	ε_{trig}	LO ratio σ(QQj)/σ(QQ), MG5 3.7.3 / 2.3.3	—	0.096 ± 0.001 / 0.096 ± 0.001	0.10
	ε_{fid}	production	62 / 186	0.333 ± 0.035	0.28
	ε_{fid}	LO sample B, Λ = 4 keV (2118 events)	636 / 2118	0.300 ± 0.010	0.28
	ε_{reco}	production, SCT z none	50 / 62	0.806 ± 0.050	0.72

Benchmark	Efficiency	Sample / setting	k / N	ε ± σ	Paper
	ε_{reco}	production, SCT z 580 μm	14 / 62	0.226 ± 0.053	0.72
	ε_{reco}	production, SCT z 1.16 mm	10 / 62	0.161 ± 0.047	0.72
	ε_{reco}	LO sample B, SCT z none	514 / 636	0.808 ± 0.016	0.72
	ε (Eq. 12)	production, SCT z none		0.0478 ± 0.0066	0.0202*
	ε (Eq. 12)	production, SCT z 580 μm		0.0134 ± 0.0036	0.0202*

* product of the paper's Table I entries (the paper does not quote ε). The relative σ of a product uses $\sigma_{rel}^2 = \Sigma\,(1-\varepsilon_i)/k_i$.



What it shows. Every efficiency of Table I at the two benchmarks: circles are the production samples, squares the EW LO sample B and the LO cross-section ratios; dashes are the paper.

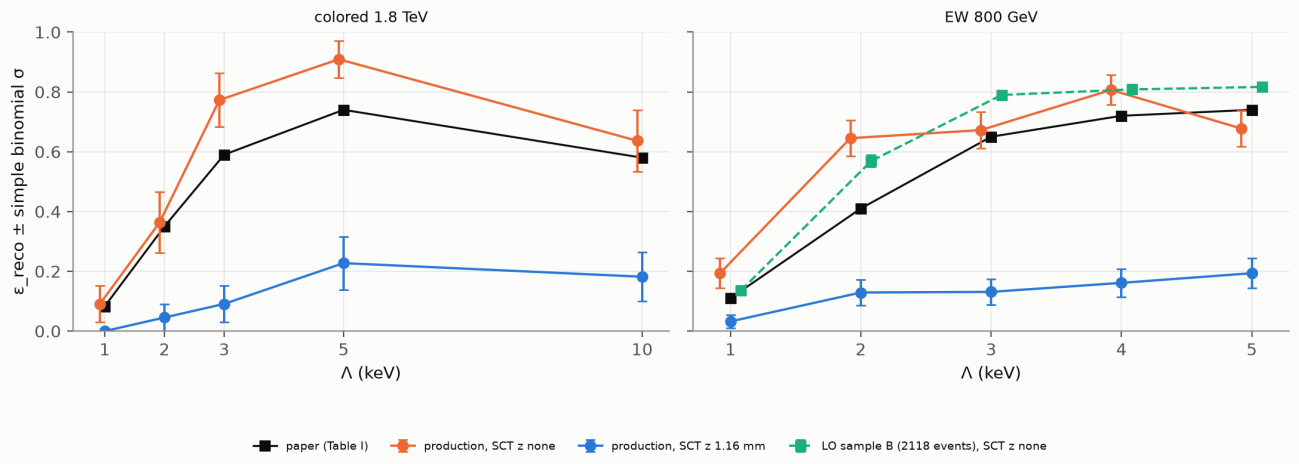
Mechanism. ε_{trig} is the fraction of pairs recoiling against initial-state radiation with $p_T(Q\bar{Q}) > 200$ GeV; the matched + showered sample has a larger fraction than the LO cross-section ratio (0.178 vs 0.096 for EW). ε_{fid} is geometry and the 25 ns cut (heavy, slow pairs). ε_{reco} is set by Δs relative to the 0.01 cm cut, which the SCT z smearing dominates.

Comparison. Colored ε_{trig} and both ε_{fid} agree with the paper. EW ε_{trig}: production 0.178, LO ratio 0.096, paper 0.10. The EW LO sample also gives $\varepsilon_{fid} = 0.300$, closer to 0.28 than the production 0.333. ε_{reco} agrees only without SCT z smearing; 580 μm and 1.16 mm give values far below the paper.

Limitation. 22 colored fiducial events (σ ≈ 0.10 on ε_{reco}). The LO sample is still running (2118/3000 events); its showered version is pending.

2.3 The Λ column of Table I (ε_{reco}, first crossing)

Benchmark	Λ (keV)	Paper	Production, none: k / N	ε ± σ	Production, 1.16 mm	EW LO sample B, none: k / N	ε ± σ
Colored	1	0.083	2 / 22	0.09 ± 0.06	0 (95% UL 0.13)	—	—
	2	0.35	8 / 22	0.36 ± 0.10	0.05 ± 0.04	—	—
	3	0.59	17 / 22	0.77 ± 0.09	0.09 ± 0.06	—	—
	5	0.74	20 / 22	0.91 ± 0.06	0.23 ± 0.09	—	—
	10	0.58	14 / 22	0.64 ± 0.10	0.18 ± 0.08	—	—
EW	1	0.11	12 / 62	0.19 ± 0.05	0.03 ± 0.02	85 / 626	0.136 ± 0.014
	2	0.41	40 / 62	0.65 ± 0.06	0.13 ± 0.04	360 / 633	0.569 ± 0.020
	3	0.65	41 / 61	0.67 ± 0.06	0.13 ± 0.04	503 / 637	0.790 ± 0.016
	4	0.72	50 / 62	0.81 ± 0.05	0.16 ± 0.05	514 / 636	0.808 ± 0.016
	5	0.74	42 / 62	0.68 ± 0.06	0.19 ± 0.05	521 / 638	0.817 ± 0.015



What it shows. ε_{reco} versus the string tension Λ. The oscillation amplitude scales as $1/\Lambda^2$ (Eq. 1), so low Λ means wide strips and many hits off the plane; high Λ means hits merging.

Mechanism. At low Λ the plane finder loses events through the Δw seed and hit-multiplicity cuts (Section 2.4); the rise with Λ follows the amplitude. With SCT z 1.16 mm, Δs ≈ 100–200 µm exceeds the 0.01 cm cut at every Λ, so the efficiency stays low.

Comparison. Colored without SCT z smearing follows the paper within about 2σ except at 3 and 5 keV (higher than the paper). EW is above the paper at 1–2 keV in both the production and the LO sample (2 keV: 0.65 production, 0.57 LO, paper 0.41). The LO sample changes the trigger and fiducial efficiencies but not this excess.

Limitation. The same events enter every Λ point, so the points of one curve are correlated.

2.4 Where events are lost (step-by-step cut-flow, D26)

Step	Colored none	Colored 580 µm	EW none	EW 580 µm
fiducial	22	22	62	62
layer-8 seed pair	19 (−3)	19 (−3)	62	62
layer-7 seed pair	19	19	62	62
seed Delta_s < 0.05	19	19	62	61 (−1)
seed Delta_w < 1	18 (−1)	18 (−1)	62	61
seed x.n3 > 0	18	18	62	61
plane found (iterative fit, Delta_w < 1)	18	18	62	61
hit multiplicity	8 (−10)	9 (−9)	50 (−12)	45 (−16)
Delta_s < 0.01 (signal region)	8	3 (−6)	50	14 (−31)

The last row is the ε_{reco} numerator; the denominator is the first row (fiducial). Without SCT z smearing most losses are in the hit-multiplicity step; with 580 µm the Δs cut removes most events.

3. Figure 2: angular momentum and quirk separation

3.1 Equations used (the paper's model, solved numerically)

Eq. 3 (straight string, Nambu–Goto with massive ends): $d(m\gamma\mathbf{v})/dt = -\Lambda^2 \left[\sqrt{(1-v_\perp^2)}\,\hat{s} + (v_\parallel/\sqrt{(1-v_\perp^2)})\,\mathbf{v}_\perp \right] + q\,\mathbf{v} \times \mathbf{B}$

with $v_\parallel = \mathbf{v}\cdot\hat{s}$ (signed), $\mathbf{v}_\perp = \mathbf{v} - v_\parallel\hat{s}$, $\hat{s}$ the unit vector from the partner to the quirk, $T = \Lambda^2$ the string tension, $B = 2$ T along z. Material effects are neglected, as in the paper ("We therefore neglect this effect in our simulations"). Integration: RK4, 3000 steps per period for Fig. 2.

Eq. 4 ($B = 0$, one quirk's distance from the CM): $d_{cm}(t) = (m/\Lambda^2)\left[\gamma - \sqrt{(1 + (\Lambda^2 t/m - v\gamma)^2)}\right]$, maximum $(\gamma - 1)\,m/\Lambda^2$; the plotted $d = |\mathbf{x}_1^* - \mathbf{x}_2^*| = 2\,d_{cm}$

$L = (\mathbf{x}_1^* - \mathbf{x}_2^*) \times (\mathbf{p}_1^* - \mathbf{p}_2^*) / 2$ in the initial CM frame at equal CM time; r = transverse radius of the CM

Eq. 5 (CM frame): $\mathbf{E}^* = \gamma_{cm}\,\mathbf{v}_{cm} \times \mathbf{B}$, torque $\boldsymbol{\tau} = q\,\mathbf{d} \times \mathbf{E}^*$ (for charges $\pm q$)

3.2 Appendix A of arXiv:2002.07503 and what it predicts

Li, Li, Pei and Zhang solve Eq. 3 with a uniform magnetic field analytically in the CM frame. In the time variable $g = \Lambda^2 t_c/(2m\rho)$ ($\rho = p^*/m$) the quirks meet ($d = 0$) at every integer g, and the motion between meetings is the $B = 0$ solution $r(g) = (m/\Lambda^2)[\sqrt{(1+\rho^2)} - \sqrt{(1+\rho^2(1-2g)^2)}]$ plus small out-of-plane terms $z^{\pm}(g)$. Appendix A states:

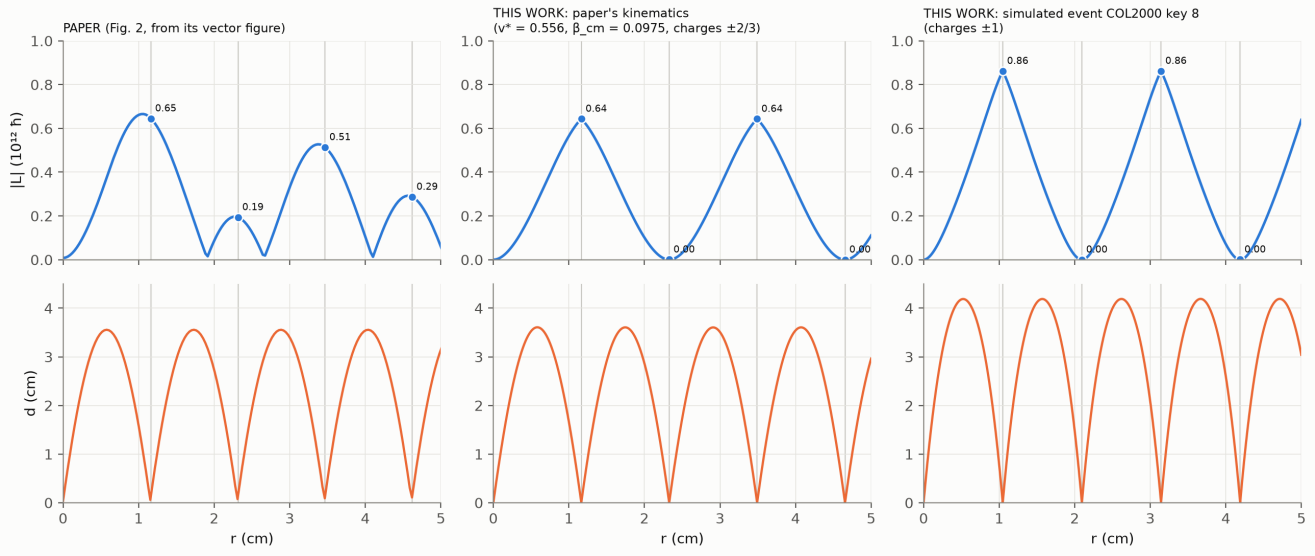
> "the average torque on the system of two quirks is zero when g is increased from 2n to 2n + 2 such that the total angular momentum of the quirk pair is also invariant at g = 2n … two quirks meet each other and have opposite velocity at g = 2n in the CoM frame."

Reason: between g = 2n and 2n + 1 quirk 1 is on one side (**d** along $+\hat{s}$), between 2n + 1 and 2n + 2 on the other (**d** along $-\hat{s}$). The electric field $\mathbf{E}^*$ is constant, so $\boldsymbol{\tau} = q\mathbf{d}\times\mathbf{E}^*$ has the same size and opposite sign in the two half-oscillations, and ΔL over a full oscillation vanishes. With L = 0 at production, |L| is therefore zero at every second meeting (g = 2, 4, …) and equal at every odd meeting. A crossing here means a closest approach, a minimum of d ($d \to 0$ up to the classical impact parameter).

Half-oscillation (constructed event, 10^{12} ħ)	1	2	3	4
change of L along its axis (trajectory)	+0.643	-0.643	+0.643	-0.643
EM torque $\int q\mathbf{d}\times\mathbf{E}^* dt$ (Eq. 5 form)	+0.556	-0.556	+0.556	-0.556
string term ($v_\parallel v_\perp$ of Eq. 3)	+0.087	-0.087	+0.087	-0.087
paper's Fig. 2 (signed, from its curves)	+0.645	-0.839	+0.707	-0.801

The string term is part of Eq. 3: at B = 0 the quirks' |L| drops to $0.8312 = 1/\gamma^*$ at the turning points and returns at the meetings, independently of v_{cm} (0, 0.0975, 0.3), and the same 0.643 is obtained when the event is integrated directly in the CM frame (D26, D21).

Fig. 2 quantity	Paper	This work: paper's kinematics (±2/3)	This work: COL2000 key 8 (±1)
d max (cm)	3.56	3.61	4.19
spacing of closest approaches in r (cm)	1.153	1.163	1.047
|L| at closest approaches 1–4 (10^{12} ħ)	0.65 / 0.19 / 0.51 / 0.29	0.64 / 0.00 / 0.64 / 0.00	0.86 / 0.00 / 0.86 / 0.00
r where |L| = 0 (cm)	1.91 / 2.66 / 4.10	2.33 / 4.65	2.09 / 4.18



What it shows. |L| (top) and the quirk separation d (bottom) versus the radius r of the centre of mass for the paper's event (left), an event built with the paper's kinematics (middle: $v^* = 0.556$ and $\beta_{cm} = 0.0975$ read from its d curve) and a simulated event (right). Grey lines mark the closest approaches.

Mechanism. d follows Eq. 4: each arch is one half-oscillation, $d_{max} = 2(\gamma^*-1)m/\Lambda^2 = 3.61$ cm, and arches repeat every β_{cm} × period = 1.16 cm in r. |L| is the integral of the torque $q\mathbf{d}\times\mathbf{E}^*$ (plus the string term): it rises during one arch and is

removed during the next, because **d** reverses. Hence |L| = 0 at every second closest approach and all peaks are equal — the Appendix A result.

Comparison. The paper's d (d_{max} 3.56 cm, spacing 1.153 cm) and its first |L| maximum (0.645) are matched (3.61 cm, 1.163 cm, 0.643). After that the paper's |L| does not return to zero: its half-oscillations are unequal (+0.65 / -0.84 / +0.71 / -0.80), and its zeros of |L| lie near the maxima of d.

Agreement. Good for d and for the first |L| peak; poor for the later |L| values. The difference is not reproduced by: four definitions of L (CM frame, lab midpoint, lab centre of energy, production point), 7 charge sets × 5 axes and 60 random configurations, 8 equation-of-motion variants (only a non-energy-conserving forward-Euler integration gives a non-zero second approach), string direction at equal CM time, an independent code (PreFASERQuirkSim) or the user-supplied simulator after correcting its units.

Unresolved. The paper does not state its L definition, the event's charges and momenta or its integrator; these are the questions for the authors.

3.3 Checks of the Fig. 2 machinery

Check	Result	Record
Eq. 3 written literally vs the integrator, 2000 random states	max difference 7×10^{-16} Λ^2	D27 (T0 tests)
Eq. 4 at B = 0 / energy $E_1 + E_2 + \Lambda^2 d$	4×10^{-11} of the amplitude / 7×10^{-12} (median)	D27
Lorentz force and units (gyroradius, p_T = 50 GeV, 2 T)	83.391 m expected, within 1.4×10^{-5}	D27
step size 3000 vs 6000 per period (Fig. 2 event)	|L| at approaches unchanged to 3×10^{-4}	D27
closest approaches with |L| < 1% of the event's |L|max, all 22 COL2000 fiducial events	132 of 276; every event	D17
independent codes: improved user simulator, PreFASERQuirkSim (CM-time string)	same pattern; peaks within 0.7%	D31, D32

4. Annihilation at the |L| ≈ 0 meetings

The paper: “whenever |L| = 0, the separation between the quirks is large, and annihilation is suppressed by a small wave-function overlap.” In the classical model (this work; arXiv:2002.07503, App. A) |L| returns to its production value at every second meeting, where d ≈ 0. This argument is therefore not supported. This does not show that annihilation occurs; the reasons are separated below by type.

Type	Statement	Basis
Simulation evidence	At the even meetings the classical |L| is below 10^{-5} of its peak but not converged: 4e+08, 6e+06, 5e+07, 5e+06 ħ at 3000, 10000, 30000, 100000 steps per period (2nd meeting). The classical impact parameter b = |L|/p* stays ≫ $1/m_Q$ = 1.1×10^{-17} cm at every resolution reached, but whether |L| would reach a few ħ cannot be decided classically.	D29 (D22 script)
Physical argument	Annihilation needs an S-wave overlap at distances ≈ $1/m_Q$; the per-meeting probability is ≈ $(\sigma v)\lvert\psi(0)\rvert^2$ × time. For an S state of $V = \Lambda^2 r$, $\lvert\psi(0)\rvert^2 = \mu\Lambda^2/2\pi$ ($\mu = m/2$), and one full oscillation lasts $4m\gamma^* v^*/\Lambda^2$, so P per oscillation ≈ $c_\sigma \alpha^2 \gamma^* v^*$, independent of m and Λ: 10^{-4}–10^{-3} for α = 0.02–0.05.	D37
Physical argument	The symmetry that returns L to zero is not exact in reality: material (2108.06748 finds that material changes L), radiation of glueballs and photons, non-uniform field. These are not in the paper's model or ours.	literature
Inherited assumption	Neither the paper nor this work simulates annihilation; Table I assumes none. arXiv:2404.13814 models annihilation only from the quirkonium ground state after radiative de-excitation.	paper; 2404.13814 Sec. 4
Quantum test (toy)	2D Schrödinger evolution of the relative coordinate with V = r − f x: the overlap at the origin returns to the field-free value at the even meetings (0.98, 0.93 for L_{peak} = 10 ħ); the incomplete return at L_{peak} = 30 ħ (0.57) is reproduced by a classical ensemble (0.52) and by a finer grid (0.565).	D37, D38

Worst case: S-wave at every second meeting, mean survival	Λ (keV)	full oscillations (median)	$\sigma v = \pi\alpha^2/m^2$, α 0.02 / 0.05	dark gluons (2404.13814), α 0.02 / 0.05
colored 1.8 TeV	2	19	0.995 / 0.970	0.999 / 0.996
colored 1.8 TeV	10	472	0.887 / 0.494	0.985 / 0.911
EW 800 GeV	4	93	0.971 / 0.835	0.989 / 0.934
EW 800 GeV	5	145	0.955 / 0.756	0.983 / 0.899

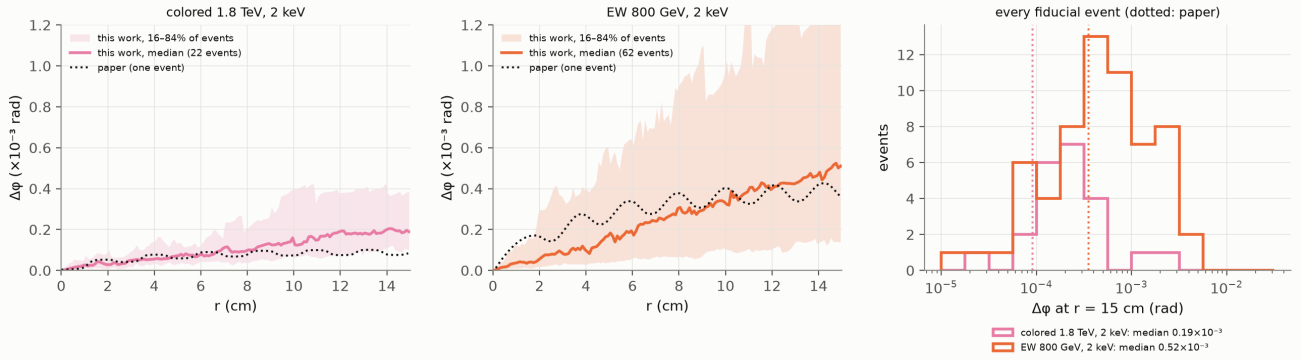
Order of magnitude (non-relativistic formula, v* ≈ 0.6); α_{dark} is not given by the paper. A worst case, not a prediction: the loss is small at the benchmarks and grows with Λ (number of oscillations ∝ Λ^2).

Statement for the record
In the classical straight-string model, |L| returns to its production value at every second meeting (this work; arXiv:2002.07503, App. A), so the paper's argument that |L| = 0 only at large separation is not supported. Whether this leads to non-negligible annihilation requires a quantum estimate of the per-meeting annihilation probability, which neither the paper nor this study provides; the worst case above bounds it from above at the Table I benchmarks.

5. Figures 3–5

Figure 3: rotation of the quirk plane

Δφ at r = 15 cm (Λ = 2 keV)	Events	median [16–84%]	Paper's event	fraction of our events below the paper
colored 1.8 TeV	22	0.19 [0.10, 0.38]×10⁻³	0.09×10⁻³	2/22 = 0.09 ± 0.06
EW 800 GeV	62	0.52 [0.14, 1.70]×10⁻³	0.36×10⁻³	24/62 = 0.39 ± 0.06



What it shows. The angle Δφ by which the plane of the two quirks' velocity vectors turns, versus r: median of all fiducial events with the 16–84% band; dotted: the paper's event.

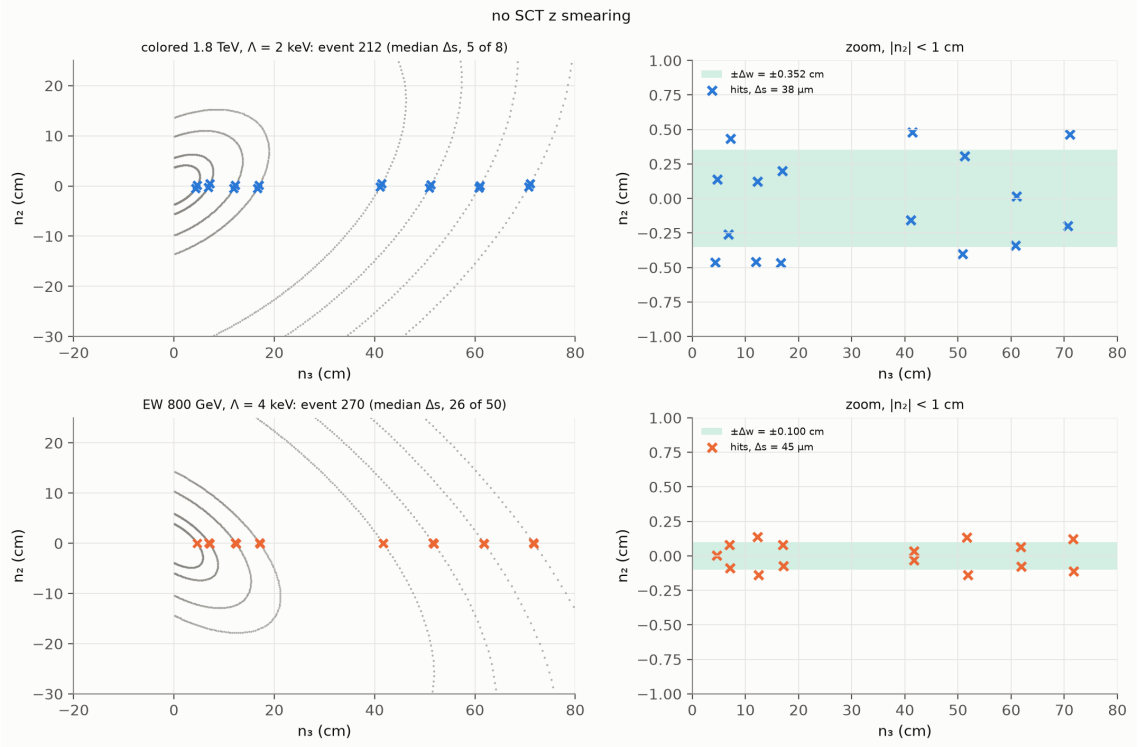
Mechanism. The torque of Eq. 5 turns the plane, but the moment of inertia $I \approx 2d^2m$ is large, so $\Delta\varphi \approx |\tau|\Delta t^2/I \approx 10^{-5}\, v_{cm}/v^2$ (paper Eq. 9). Δφ oscillates with the half-oscillations and accumulates slowly.

Comparison. The paper's single event lies inside our distribution (at its low edge for colored).

Agreement / limitation. Good: the paper's event lies within our spread. Δφ describes the plane of the velocity vectors; whether the hits stay coplanar is measured directly by the truth plane thickness of the hits (a few μm, Section 6.2). One paper event only; its kinematics are unknown.

Figure 4: hits of one signal event in the plane's coordinates

Fig. 4 example (median Δs in the signal region)	SCT z	Event	rank	Δs	Δw
Colored	1.16 mm	201	1 of 1	52 μm	0.470 cm
EW	1.16 mm	301	6 of 10	73 μm	0.079 cm
Colored	none	212	5 of 8	38 μm	0.352 cm
EW	none	270	26 of 50	45 μm	0.100 cm
paper example (colored)	—	—	—	83 μm	0.074 cm



What it shows. The hits of the signal-region event with the median Δs (no SCT z smearing), projected on the fitted plane (n_3, n_2); dotted ellipses: the layers; zoom: the strip.

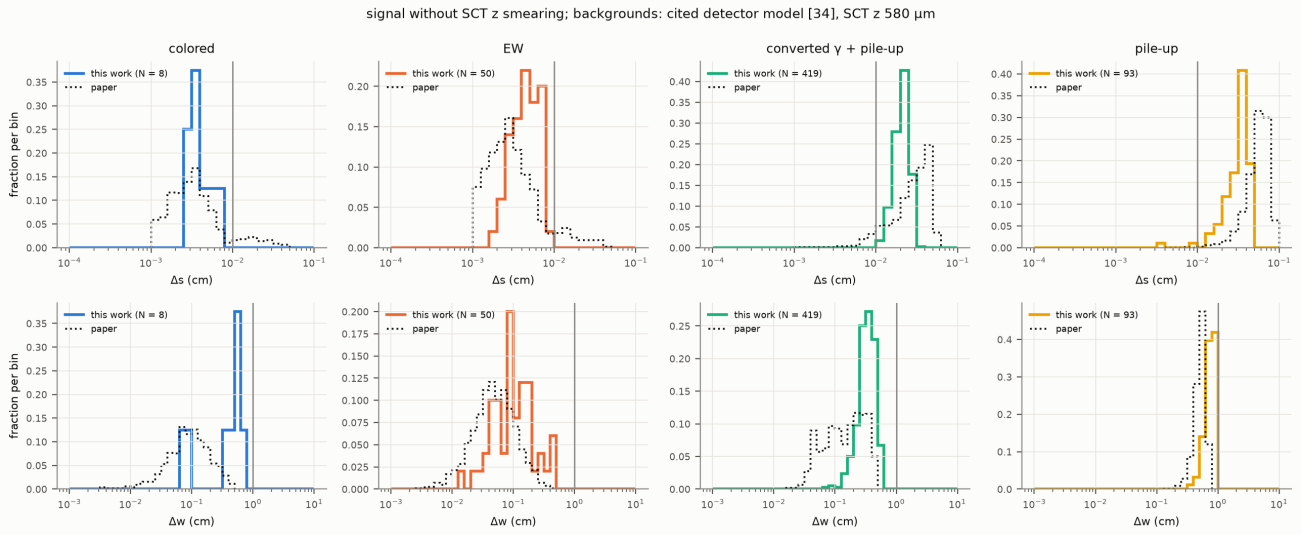
Mechanism. The two quirks oscillate about the pair's direction n_3; the hits fill a strip of full width ≈ the belt width L of arXiv:2002.07503 Eq. III.1, Δw ≈ L/√12 for a uniform strip.

Comparison. Same pattern as the paper's Fig. 4 (hits on a narrow strip along n_3). The paper's example has Δw = 0.074 cm and Δs = 83 μm; ours (fixed median rule) is wider.

Limitation. One event per panel, chosen by a fixed rule; with 1.16 mm (F29_fig4_Z1160, previous report) only one colored event passes.

Figure 5: Δs and Δw distributions

Population	SCT z	N	Δs median	paper	Δw median	paper	Δs < 0.01 cm: k / N, ε ± σ	paper
colored	none	8	38 μm	32 μm	0.482 cm	0.092 cm	8 / 8 = 1.00 ± 0.00	0.90
colored	1.16 mm	4	160 μm	32 μm	0.550 cm	0.092 cm	1 / 4 = 0.25 ± 0.22	0.90
EW	none	50	44 μm	28 μm	0.097 cm	0.048 cm	50 / 50 = 1.00 ± 0.00	0.92
EW	1.16 mm	24	110 μm	28 μm	0.092 cm	0.048 cm	10 / 24 = 0.42 ± 0.10	0.92
converted γ + pile-up	580 μm, [34]	419	212 μm	306 μm	0.340 cm	0.156 cm	0 / 419 = 0 (95% UL 0.01)	0.07
pile-up	580 μm, [34]	93	340 μm	571 μm	0.751 cm	0.525 cm	2 / 93 = 0.02 ± 0.02	0.00



What it shows. Δs (plane thickness) and Δw (strip width) of signal events after the plane and hit cuts (before the Δs cut), and of the backgrounds; unit area; dotted: the paper.

Mechanism. The true plane thickness is a few μm (Section 6.2), so Δs is set by the detector resolution: pixel rounding (14–115 μm RMS) and the SCT 34 μm rφ give ≈ 40 μm; an SCT z smearing of 0.58–1.16 mm adds ≈ 100–200 μm. Δw is set by the oscillation amplitude and the boost (belt width).

Comparison. Without SCT z smearing the signal Δs (38 / 43 μm) is close to the paper's (32 / 28 μm) and nearly all signal lies left of the cut, while backgrounds lie to the right. Δw is wider than the paper's: colored 5×, EW 2×.

Agreement / limitation. Δs: good only without SCT z smearing (with 1.16 mm: F30_fig5_Z1160, ≈ 150 μm). Δw: poor. 8 colored and 50 EW signal events.

6. Tests of Δw and of the plane thickness

6.1 Δw (strip width)

Test	Colored	EW	Paper
Δw median, production, plane + hit cut (SCT z none)	0.482 cm	0.097 cm	0.092 / 0.048 cm
Δw median, population “plane found”, before the hit and Δs cuts (authors: Fig. 5 is before the Δs and Δw cuts)	0.498 cm	0.096 cm	—
belt width L (2002.07503 Eq. III.1), median over fiducial events	1.699 cm	0.270 cm	—
Δw / L (median); uniform strip: 1/√12 = 0.289	0.280	0.350	—
Spearman correlation of Δw with L	0.74	0.96	—
Δw median, string at equal CM time instead of lab time (D40)	0.515 cm	0.093 cm	—
Δw median, EW LO sample B (2118 events), Λ = 4 keV, plane found	—	0.124 cm	—

Test	Colored	EW	Paper
EW LO sample B, showered without matching	—	pending	—

Our Δw follows the analytic belt width of our own kinematics (ratio ≈ 1/√12, strong correlation) and does not change with the population, the string convention or the LO sample. The 2–5× difference to the paper is not produced by the reconstruction; its origin (event kinematics or the plotted definition) is open.

6.2 Plane thickness without detector effects

Truth thickness (exact first crossings, no resolution)	Colored 2 keV	EW 4 keV
this work, string at equal lab time (production)	4.12 µm	13.00 µm
string at equal CM time (paper / 2002.07503 Sec. VIII)	1.89 µm	0.59 µm
analytic: 1911.02223 Eq. III.12 / 2002.07503 Eq. V.6	2.14 / 0.55 µm	0.06 / 0.12 µm

Effect of the string convention (D40)	lab time: ε_{reco} (none)	CM time	Δs median lab → CM
colored 2 keV	8/22 = 0.36 ± 0.10	8/22 = 0.36 ± 0.10	37.7 → 40.1 µm
EW 1 keV	12/62 = 0.19 ± 0.05	11/62 = 0.18 ± 0.05	37.2 → 27.8 µm
EW 2 keV	40/62 = 0.65 ± 0.06	42/62 = 0.68 ± 0.06	42.0 → 32.9 µm
EW 4 keV	50/62 = 0.81 ± 0.05	51/62 = 0.82 ± 0.05	43.1 → 37.9 µm

Our production code takes the string direction at equal lab time; the paper states the straight string in the CM frame. This raises the truth thickness (EW 13 → 0.6 µm) and lowers the EW Δs without SCT z smearing by 5–9 µm, but changes ε_{reco} by at most 2 events. It does not explain the Table I differences.

7. Backgrounds and the converted-photon upper limit

Sample	Events	Plane	+ hit mult.	Signal region	per-event rate
converted γ + μ = 50, SCT z 580 µm (cited model)	9139	3399	419	0 / 9139	0 (95% UL 0.0003)
converted γ + μ = 50, SCT z 1.16 mm, no [34] energy loss	9139	1990	48	0 / 9139	0 (95% UL 0.0003)
converted γ + μ = 50, no SCT z smearing	9139	4105	285	146 / 9139	0.0160 ± 0.0013
pile-up only, μ = 50, SCT z 580 µm	5000	—	—	0 / 5000	0 (95% UL 0.0006)
pile-up: plane with ≥ 1 hit in ≥ 4 of 8 layers (paper ~10^{-3})	5000			33 / 5000	0.0066 ± 0.0011
pile-up: plane with ≥ 1 hit in all 8 layers (paper 6×10^{-5})	5000			1 / 5000	0.00020 ± 0.00020

Converted photons at 300 fb^{-1}	σ(Z(νν)γ + j)	conversions (×1%)	signal-region fraction	expected events
this work's cross section, SCT z 580 µm	0.211 pb	633	0/9139 → 95% UL 3.3×10^{-4}	< 0.21
this work's cross section, no SCT z smearing	0.211 pb	633	146/9139 = 0.0160 ± 0.0013	10.1 ± 0.8
paper's cross section, SCT z 580 µm	1.0 pb	3000	0/9139 → 95% UL 3.3×10^{-4}	< 0.98
paper's cross section, no SCT z smearing	1.0 pb	3000	146/9139 = 0.0160 ± 0.0013	47.9 ± 3.9

Why the upper limit is the same for both cross sections. The signal-region fraction is measured per converted photon on the same 9139 simulated events. With 0 passing, its 95% upper limit is $1 - 0.05^{1/9139} = 3.3\times10^{-4}$, which depends only on the number of simulated events. The cross section enters only the next step, the expected number of conversions (300 fb^{-1} × σ × 1%): 633 or 3000, hence < 0.21 or < 0.98 events. If the background is simulated without SCT z smearing, 146 conversions pass and the expected number becomes 10–48 events: the zero-background statement relies on the SCT z smearing of the background, which is the setting the signal efficiencies do not tolerate.

8. Investigation record since 9 October

Record	Question	Result	Status
D14	are 300 RK4 steps per period enough?	hit positions not converged (median shift 0.4–0.6 mm vs 6000); signal-region counts change by ±1 (EW: 7 events flip both ways)	accepted; listed as numerical uncertainty
D17–D28	why the paper's Fig. 2 |L| differs	L definition, charges, axes, EOM variants, two independent codes, B = 0 test, CM-frame integration: our pattern is the solution of Eq. 3	verified; paper's curve unexplained
D27	are the equations implemented as written?	Eq. 3, Lorentz force, units, Eq. 4, energy: agreement 10^{-16}–10^{-5}	verified
D29, D37–D38	annihilation at |L| ≈ 0	classical |L| at even meetings < 10^{-5} of the peak (not converged); worst case small at the benchmarks; quantum toy follows the classical return	open (needs a quantum estimate)

Record	Question	Result	Status
D33–D34	literature (Setford thesis; 2002.07503; 2404.13814; 2108.06748; 1911.02223)	material neglected in the paper; App. A of 2002.07503 = our Fig. 2; EW trigger 0.10 = LO ratio (1911.02223)	integrated
D35–D36	settings, model, tests	colored quirk is a fermion; config files with strict checks; 6 physics tests pass; outputs unchanged (byte-identical)	done
D38	repeated crossings	13/22 colored and 31/62 EW fiducial events have > 2 hits on a layer (paper: “most likely ... only once”)	verified
D39–D40	string direction at equal CM time	truth thickness EW 13 → 0.6 μm; ε_{reco} within ±2 events	verified; modelling approximation listed
D41	EW sample as LO $Q\bar{Q}$ + 1 jet	ε_{trig} 0.096 (= paper 0.10), ε_{fid} 0.300 (paper 0.28); ε_{reco} unchanged	running

9. Unresolved differences and next steps

- **EW ε_{reco} at Λ = 1–2 keV** above the paper (2 keV: production 0.65 ± 0.06, LO 0.57 ± 0.02, paper 0.41). Not removed by the LO sample, the string convention or the step size.
- **SCT z resolution.** The authors' stated resolution (580 μm) gives ε_{reco} far below Table I; only no smearing reproduces it. Question for the authors: which coordinate and value.
- **Signal Δw** 2× (EW) and 5× (colored) the paper's; our Δw follows the analytic belt width of our kinematics.
- **Fig. 2**: the paper's unequal half-oscillations; questions: L frame and point, whether L includes the string's share, the event's charges and momenta.
- **Annihilation**: a quantum per-meeting estimate is needed to turn the worst case into a number.
- **Next:** finish D41 (3000 events) and its showered sensitivity; ask the authors; pile-up 4-of-8 plane rate (6.6×10^{-3} vs $\sim10^{-3}$) remains open.

10. Provenance

Result	Data file(s)	Script
production counts, Table I	tables/all_crossings_per_event.csv; quirk_2000_event_fresh_regeneration_v1/events/*/matched_events_full.csv	P18_all_crossings.py
Λ column	tables/lambda_scan_summary.csv, lambda_scan_z1160_summary.csv	P20_lambda_scan.py, P20b
EW LO trigger and sample B	inputs_D41/ (MG5 logs, banners, LHE, shower CSV); logs/D41_mg5_373_lo_events.jsonl	D41_ew_lo_ratio_pipeline.py
backgrounds	logs/P10_chunks_*, P21_chunks_REF34_Z580, P8_chunks_LITERAL	P10, P21
Fig. 2	tables/fig2_final/*.csv; tables/D16–D22, D17_fig2_torque_decomposition.txt	F38_report_2026_10_15_figs.py; D16–D25
Fig. 3, 4, 5	FINAL_FIGURES/F30_fig3, F29_fig4_*, F30_fig5_* (source CSVs); tables/paper_fig5_vector.json	F30_report_figs.py, F29
annihilation	tables/D27_annihilation_A.csv, D27/D28 txt, D22 txt	D27, D28, D22
width / thickness	tables/D29_li_pei_formulas.csv, D32_cm_time_string_reco.csv, fig5_author_population.csv	D29, D32, P24
all numbers of this report	—	R_data_2026_10_15.py (assertions), R_report_2026-10-15.py

Settings: config/detector.yaml, reconstruction.yaml, benchmarks.yaml (checked by T1_config_check.py); physics tests T0_physics_tests.py. Decision log entries D17–D41.

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